

Cell-type-specific fluorescent tagging of endogenous target proteins reveals synaptic enrichment and dynamic regulations of dopamine receptors

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Shun Hiramatsu, Kokoro Saito, Shu Kondo, Hidetaka Katow, Nobuhiro Yamagata, Chun-Fang Wu, Hiromu Tanimoto 

Graduate School of Life Sciences, Tohoku University • Department of Biological Science and Technology, Faculty of Advanced Engineering, Tokyo University of Science • Department of Cell Biology, New York University • Faculty and Graduate School of Engineering Science, Akita University • Department of Biology, University of Iowa

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Abstract

Dopamine can play opposing physiological roles depending on the receptor subtype. In the fruit fly *Drosophila melanogaster*, *Dop1R1* and *Dop2R* encode the D₁- and D₂-like receptors, respectively, and are reported to oppositely regulate intracellular cAMP levels. Here, we profiled the expression and subcellular localization of endogenous Dop1R1 and Dop2R in specific cell types in the mushroom body circuit. For cell-type-specific visualization of endogenous proteins, we employed reconstitution of split-GFP tagged to the receptor proteins. We detected dopamine receptors at both presynaptic and postsynaptic sites in multiple cell types. Quantitative analysis revealed enrichment around the active zones, particularly for Dop2R. The presynaptic localization of Dop1R1 and Dop2R in dopamine neurons suggests dual feedback regulation as autoreceptors. Furthermore, we discovered a starvation-dependent, bidirectional modulation of the presynaptic receptor expression in the PAM and PPL1 clusters, two distinct subsets of dopamine neurons, suggesting regulation of appetitive behaviors. Our results highlight the significance of the co-expression of the two antagonizing dopamine receptors in the spatial and conditional regulation of dopamine responses in neurons.

eLife assessment

This study uses state-of-the-art methods to label endogenous dopamine receptors in a subset of *Drosophila* mushroom body neuronal types. The authors report that DopR1 and Dop2R receptors, which have opposing effects in intracellular cAMP, are present in axons termini of Kenyon cells, as well as those of two classes of dopaminergic neurons that innervate the mushroom body indicative of autocrine modulation by dopaminergic neurons. Additional experiments showing opposing effects of starvation on DopR1 and Dop2R levels in mushroom body neurons are consistent with a role for dopamine receptor levels increasing the efficiency of learned food-odour associations in starved flies. Supported by **solid** data, this is a **valuable** contribution to the field.

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Introduction

Neurotransmitters typically have multiple cognate receptors, and they may recruit different second messenger systems. Therefore, the expression and localization of receptor subtypes are critical for determining cellular responses to neurotransmitter inputs. The dopaminergic system offers an ideal *in-vivo* study case to this end, as it regulates a wide array of physiological functions through combinations of different receptor subtypes. In mammals, D₁-like receptors are coupled to G_{α_s}, thereby activating adenylate cyclase upon ligand binding, whereas G_{α_i}-coupled D₂-like receptors inhibit cyclase activity (Beaulieu & Gainetdinov, 2011 [DOI](#)). Four dopamine receptors have been identified in *Drosophila*: *Dop1R1*, *Dop1R2*, *Dop2R*, and *DopEcR* (K. A. Han et al., 1996 [DOI](#); Hearn et al., 2002 [DOI](#); Srivastava et al., 2005 [DOI](#); Sugamori et al., 1995 [DOI](#)). *Dop1R1* and *Dop2R* correspond to the D₁- and D₂-like receptors, respectively (Hearn et al., 2002 [DOI](#); Sugamori et al., 1995 [DOI](#)). *Dop1R2* and *DopEcR* are invertebrate-specific and have been reported to recruit different second messenger systems (K. A. Han et al., 1996 [DOI](#); Srivastava et al., 2005 [DOI](#)). Intriguingly, recent data of single-cell RNA-seq and transgenic expression profiling revealed that the expression of these dopamine receptors is highly overlapping in the fly brain (Croset et al., 2018 [DOI](#); Davie et al., 2018 [DOI](#); Dolan et al., 2019 [DOI](#); Kondo et al., 2020 [DOI](#)), unlike the spatially segregated expression of the D₁- and D₂-like receptors in vertebrate brains (Gerfen & Surmeier, 2011 [DOI](#)). Considering the opposing physiological roles of the *Dop1R1* and *Dop2R*, their protein localization especially in those cells where they are coexpressed should be critical to determine the responses to the dopamine inputs.

Drosophila mushroom bodies (MB) have long served as a unique dopaminergic circuit model to study adaptive behaviors, such as associative memory. MB-projecting neurons and their connections have been systematically described at both mesoscopic and ultrastructural resolutions (Aso, Hattori, et al., 2014; Li et al., 2020 [DOI](#); Takemura et al., 2017 [DOI](#); Tanaka et al., 2008 [DOI](#)). Kenyon Cells (KCs), the major MB intrinsic neurons, encode a variety of sensory information (Honegger et al., 2011 [DOI](#); Turner et al., 2008 [DOI](#); Vogt et al., 2014 [DOI](#)). Intriguingly, each KC receives synaptic inputs from different dopaminergic projections in multiple spatially segmented compartments along its axon in the MB lobe (Aso, Hattori, et al., 2014; Tanaka et al., 2008 [DOI](#)). MB-projecting dopamine neurons (DANs) originate from the three cell clusters (Mao & Davis, 2009 [DOI](#); Nässel & Elekes, 1992 [DOI](#)). DANs in the protocerebral posterior lateral 1 (PPL1) cluster project to the vertical lobes and the peduncle of the MB, and they control different aspects of associative memory (Aso et al., 2010 [DOI](#), 2012 [DOI](#); Aso & Rubin, 2016 [DOI](#); Claridge-Chang et al., 2009 [DOI](#); Krashes et al., 2009 [DOI](#); Mao & Davis, 2009 [DOI](#); Masek et al., 2015 [DOI](#); Riemensperger et al.,

2005 [2005](#); Takemura et al., 2017 [2017](#); Vogt et al., 2014 [2014](#)). The protocerebral anterior medial (PAM) cluster, the largest DAN cluster, mostly projects to the medial lobes of the MB, and many PAM neurons are involved in reward processing (Burke et al., 2012 [2012](#); Felsenberg et al., 2018 [2018](#); Huetteroth et al., 2015 [2015](#); Ichinose et al., 2021 [2021](#); Lin et al., 2014 [2014](#); Liu et al., 2012 [2012](#); Yamagata et al., 2015 [2015](#), 2016 [2016](#)). DANs in PPL2ab project to the MB calyx and control the conditioned odor response (Boto et al., 2019 [2019](#)). In addition to the variety of the dopamine sources, KCs express all four dopamine receptor subtypes (Deng et al., 2019 [2019](#); Kondo et al., 2020 [2020](#)). Given the multitude of modulatory effects of dopamine in the MB (Berry et al., 2018 [2018](#); Cohn et al., 2015 [2015](#); Handler et al., 2019 [2019](#)), receptor localization in KCs provides important information for interpreting such functional diversity.

The projections and synapses of the neurons in the MB circuit are tightly intertwined (Li et al., 2020 [2020](#); Takemura et al., 2017 [2017](#)). Therefore, conventional immunohistochemical approaches using light microscopy do not allow identification of cells from which immunoreactive signals originate. Precise determination of their subcellular localization requires visualization of the proteins of interest only in the target MB neurons (Fendl et al., 2020 [2020](#)). Employing the CRISPR/Cas9-mediated split-GFP tagging strategy (D. Kamiyama et al., 2016 [2016](#); R. Kamiyama et al., 2021 [2021](#); Kondo et al., 2020 [2020](#)), we profiled the spatial distribution of endogenous Dop1R1 and Dop2R proteins in KCs, the PAM, and the PPL1 DANs.

Results

Co-expression of *Dop1R1* and *Dop2R* genes in the adult *Drosophila* brain

To compare the expression of different dopamine receptor genes in detail, we used fluorescent reporter knock-ins of the endogenous *Dop1R1* and *Dop2R* genes (Kondo et al., 2020 [2020](#)). Both lines labelled many neuropils including the MB (**Figure 1A** [2020](#) and **1B** [2020](#)), and the overlapping expression of these receptors is consistent with our previous quantification of GAL4-positive cells for both genes (58,049 and 68,528 for *Dop1R1* and *Dop2R*, respectively, out of 118,331 brain cells; Kondo et al., 2020 [2020](#)). Double labelling of *Dop1R1* and *Dop2R* with *T2A-LexA* and *T2A-GAL4*, respectively, revealed cells with overlapping and differential expression (**Fig. 1D-G** [2020](#); see also Kondo et al., 2020 [2020](#)). The PAM cluster of DANs expressed both *Dop1R1* and *Dop2R* (**Figure 1D** [2020](#)). On the other hand, most of the DANs in the PPL1 cluster strongly expressed *Dop2R*, but *Dop1R1* only weakly (**Figure 1E** [2020](#)). We further found that *Dop1R1*, but not *Dop2R*, was highly expressed in the ring neurons of the ellipsoid body (**Figure 1F** [2020](#); Hanesch et al., 1989 [1989](#)) and the neuropil ensheathing glia (**Figure 1G** [2020](#); Awasaki et al., 2008 [2008](#)). In conclusion, *Dop1R1* and *Dop2R* genes are co-expressed in the PAM neurons, KCs and a subset of the PPL1 neurons.

Quantification of subcellular enrichment of endogenous proteins in target cells

The overlapping expression of *Dop1R1* and *Dop2R* genes prompted us to examine the subcellular localization of these receptor proteins. To elucidate the localization of these broadly expressed receptors (**Figure 1** [2020](#)), we took advantage of split-GFP tagging of endogenous proteins (D. Kamiyama et al., 2016 [2016](#); Kondo et al., 2020 [2020](#)). By adding GFP₁₁ tags to the C-termini of the *Dop1R1* and *Dop2R* proteins, their intracellular distribution can be visualized specifically in cells expressing GFP₁₋₁₀ through split-GFP reconstitution (**Figure 2A** [2020](#)).

To quantify the subcellular enrichment of the receptors, we devised Localization Index (LI). Briefly, LI is the normalized ratio of the reconstituted GFP (rGFP) signal to the reference membrane marker signal (CD4::tdTomato). If the target and reference proteins had the identical

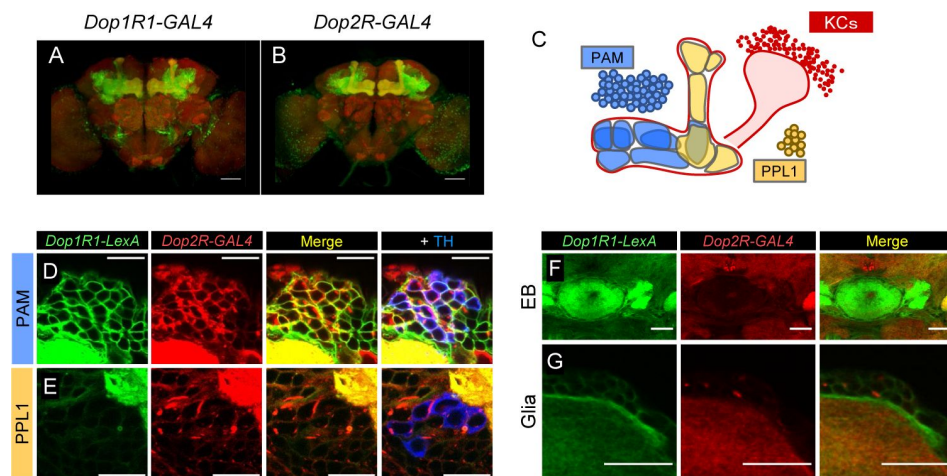


Figure 1.

Co-expression of *Dop1R1* and *Dop2R* genes in adult *Drosophila* brain.

(A and B) The expression of *Dop1R1-T2A-GAL4* and *Dop2R-T2A-GAL4* visualized by *UAS-mCD8::GFP* (green) The neuropil was stained by using *Brp::SNAP* (red). Maximum-intensity projections of the whole brain.

(C) Schematic of the KCs and the MB-innervating dopamine neurons from the PAM and PPL1 clusters.

(D-G) Double labelling of *Dop1R1* and *Dop2R* gene expression visualized by *Dop1R1-T2A-LexA/lexAop-rCD2::GFP* (green) and *Dop2R-T2A-GAL4/UAS-CD4::tdTomato* (red). Dopamine neurons were immunostained with anti-TH antibody (blue). Single optical sections. Cell bodies of the PAM cluster (D) and the PPL1 cluster (E). The ellipsoid body (F) and ensheathing glia (G). Scale bars, 50 μ m (A and B), 5 μ m (D-G).

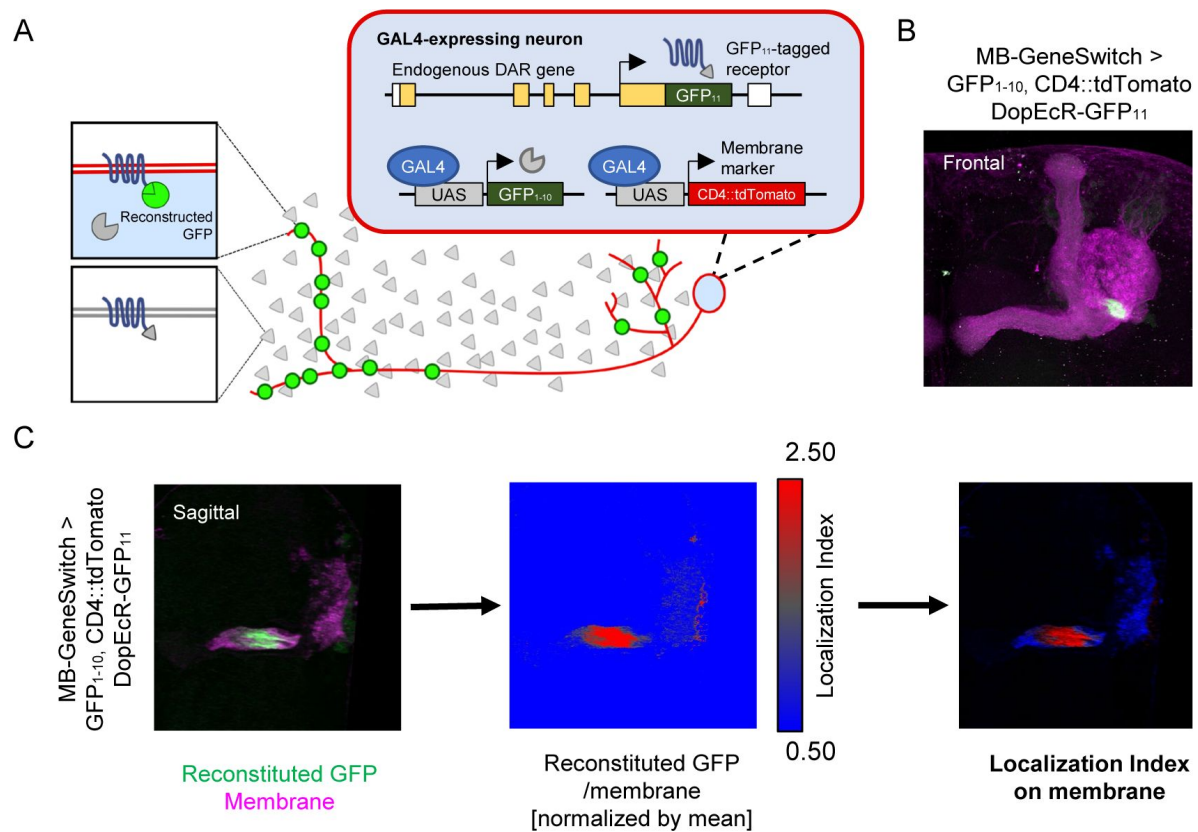


Figure 2.

Cell type-specific visualization of endogenous proteins with GFP11 tag.

(A) Principle of cell type specific fluorescent labelling of target proteins by GFP₁₁ tag. GFP₁₋₁₀ and membrane marker CD4::tdTomato are expressed in the target cells by GAL4/UAS system. In the target cells, reconstitution of GFP occurs on the endogenous proteins tagged with GFP₁₁.

(B) As an example, DopEcR::GFP₁₁ is visualized in KCs using *MB-GeneSwitch*, a ligand-inducible GAL4 driver. A merged image of reconstituted GFP (green) and cellular membrane visualized by *UAS-CD4::tdTomato* (magenta). Maximum intensity projection of the whole left MB.

(C) The workflow for visualizing subcellular protein enrichment by localization index (LI). A single sagittal section of the MB calyx and peduncle is shown. The ratio of reconstituted GFP to membrane signal in the left image is calculated and normalized by the mean intensity of all voxels to provide LI. In the middle image, LI is color-coded so that red represents local receptor enrichment. In the right image, the LI color is mapped on the membrane signal.

distribution, LI would be 1 everywhere. More or less concentrations of rGFP result in the LI larger or smaller than 1, respectively (**Figure 2C**; see also Methods for detail). LI was visualized as the color of the membrane signals (**Figure 2C**). As proof of principle, mapping the LI of DopEcR-rGFP signals in KCs highlighted enrichment in the proximal peduncle (**Figure 2B, C**), which is consistent with the previous report (Kondo et al., 2020). This representation thus visualizes the subcellular enrichment of the targeted receptors in the plasma membrane of GAL4-expressing cells.

Partial colocalization of Dop1R1 and Dop2R proteins

First, we compared the localization of Dop1R1 and Dop2R proteins in KCs, where both receptor genes were highly expressed (**Figure 1D**). Both receptors were distributed throughout KC membranes with predominant enrichment in the MB lobes (**Figure 3B-E**). This lobe enrichment was more pronounced for Dop2R than Dop1R1 (**Figure 3F** and **3G**). LI measurements indicated that Dop2R was approximately twice as enriched in the lobe as the average density across the whole neuron, while the lobe enrichment of Dop1R1 was about 1.5 times the average (**Figure 3H**).

As KCs have major presynaptic sites in the MB lobes, we examined receptor localization to the active zones (AZ) using antibodies against Brp, an AZ scaffolding protein (Wagh et al., 2006). Interestingly, Dop1R1 and Dop2R proteins in KCs were tightly associated with the Brp puncta in the lobe (**Figure 4A** and **4B**). The distribution of Dop2R, compared to Dop1R1, showed a greater association, as measured by the overlap of rGFP and Brp signals (**Figure 4C**). This was consistent with the higher lobe enrichment of Dop2R (**Figure 3H**). The AZ localization (Christiansen et al., 2011) of Dop1R1 and Dop2R was similar in the calyx (**Figure 4 – figure supplement 1**), although the identity of presynaptic neurons is less clear.

To better resolve the presynaptic localization of Dop1R1 and Dop2R, we turned to “giant” *Drosophila* neurons differentiated from cytokinesis-arrested neuroblasts in culture (Wu et al., 1990). These “giant” neurons exhibit action potential firing patterns and neurotransmission similar to those described in mature neurons (Saito & Wu, 1991; Yao and Wu, 2001; Yao et al., 2000). Due to the large size of the cells, this system has advantages for investigating the microanatomy of neurons (Saito & Wu, 1991; Wu et al., 1990; Yao et al., 2000). In the giant neurons from the *Brp::SNAP* embryos (Kohl et al., 2014), we found punctate aggregation of Brp in the terminals of neurites, indicating the proper assembly of the AZ cytomatrix (**Figure 4D** and **4E**). We confirmed that the giant neurons express Venus-tagged Dop1R1 and Dop2R (**Figure 4F-4I**). Strikingly, double labelling of these receptors with Brp::SNAP showed the association, but not overlap, with the Brp puncta, suggesting perisynaptic receptor accumulation (**Figure 4F-4I**). A closer investigation revealed a tighter association between Dop2R and Brp (**Figure 4G** and **4I**). Presynaptic enrichment of these receptors in the giant neurons was strikingly similar to their localization in KCs, further substantiating presynaptic dopaminergic modulation.

D₂-like receptors in mammals are expressed in DANs and act as autoreceptors, which mediate feedback signals by receiving neurotransmitters released from the neuron on which the receptors reside (Ford, 2014). In *Drosophila*, multiple dopamine receptor genes are expressed in DANs (**Figure 1**; Deng et al., 2019; Aso et al., 2019; Kondo et al., 2020), but it is unclear if these receptors function as autoreceptors. We therefore examined the subcellular localization of Dop1R1 and Dop2R in the PAM cluster DANs with a particular focus on their presynaptic terminals. The PAM neurons are polarized; presynaptic proteins are abundantly enriched in the MB lobe and barely detected in dendrites (**Figure 5A-5C**). We visualized Dop1R1 and Dop2R proteins in the PAM cluster DANs using *R58E02-GAL4*, and both were localized to the terminals in the MB lobes (**Figure 5A** and **5B**). Representation of LI for Dop1R1 and Dop2R in the PAM neurons again showed stronger presynaptic enrichment of Dop2R than Dop1R1 (**Figure 5F** and **5G**). Dop2R was strongly enriched at β¹ compartment showing significantly higher LI than Dop1R1 (**Figure**

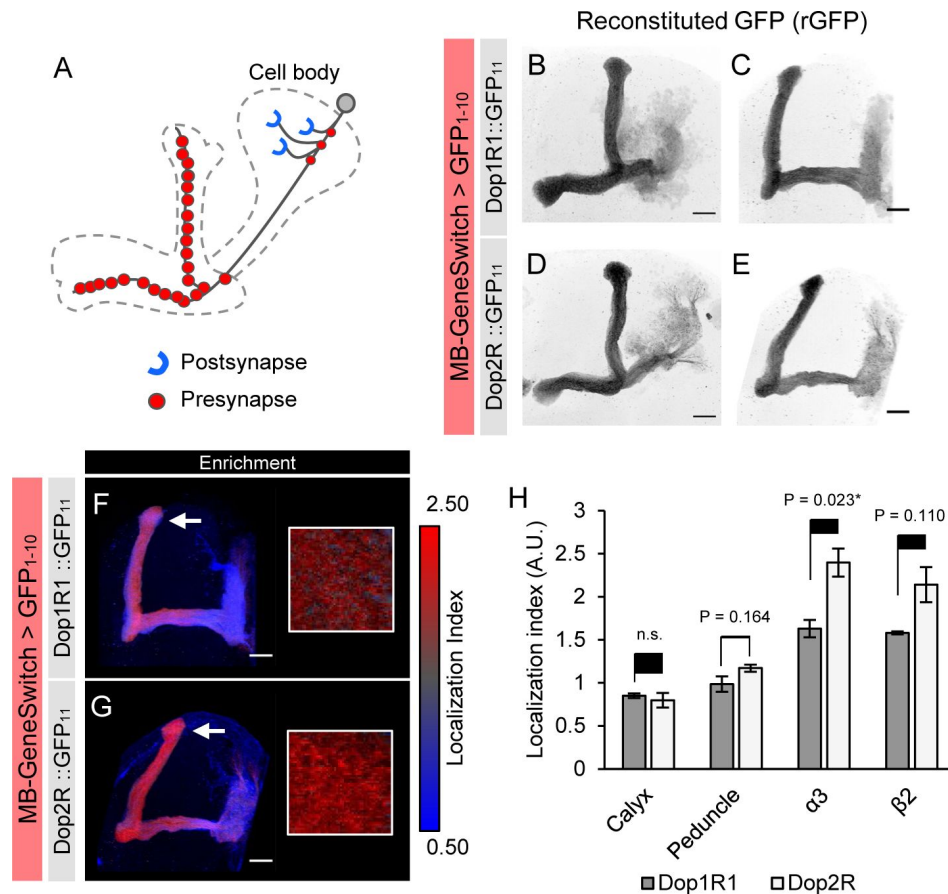


Figure 3.

Subcellular localization of Dop1R1 and Dop2R in the Kenyon Cells.

Subcellular localization of Dop1R1 and Dop2R in the KCs is visualized by GFP₁₁-tag. *MB-GeneSwitch* was used to express GFP₁₁ and CD4::tdTomato in the KCs. Receptors in $\alpha\beta$ KCs were visualized after 12 hours of RU486 feeding (see Materials and Methods).

(A) Schematic of the distribution of presynapses and postsynapses in the KCs.

(B-G) Enrichment of Dop1R1 and Dop2R in the MB lobe. Frontal views of the MB (B and D) and sagittal views of the MB (C, E, F and G) of the maximum-intensity projections of the whole left MB are shown. Reconstituted GFP signals for both Dop1R1:: and Dop2R::GFP₁₁ distributed throughout the MB lobe and the calyx (B-E). Visualization by LI showed more pronounced enrichment of Dop2R (G) than Dop1R1 (F) in the lobe as exemplified in the tip (arrows, 10 μ m square, magnified in the insertion). Scalebars, 20 μ m (B-G).

(H) Mean LI for Dop1R1 and Dop2R in the calyx, the peduncle, the tip of α and β lobes. Student's t test was performed to compare LI of Dop1R1 and Dop2R in each region (N = 3). Error bars; SEM. * p < 0.05.

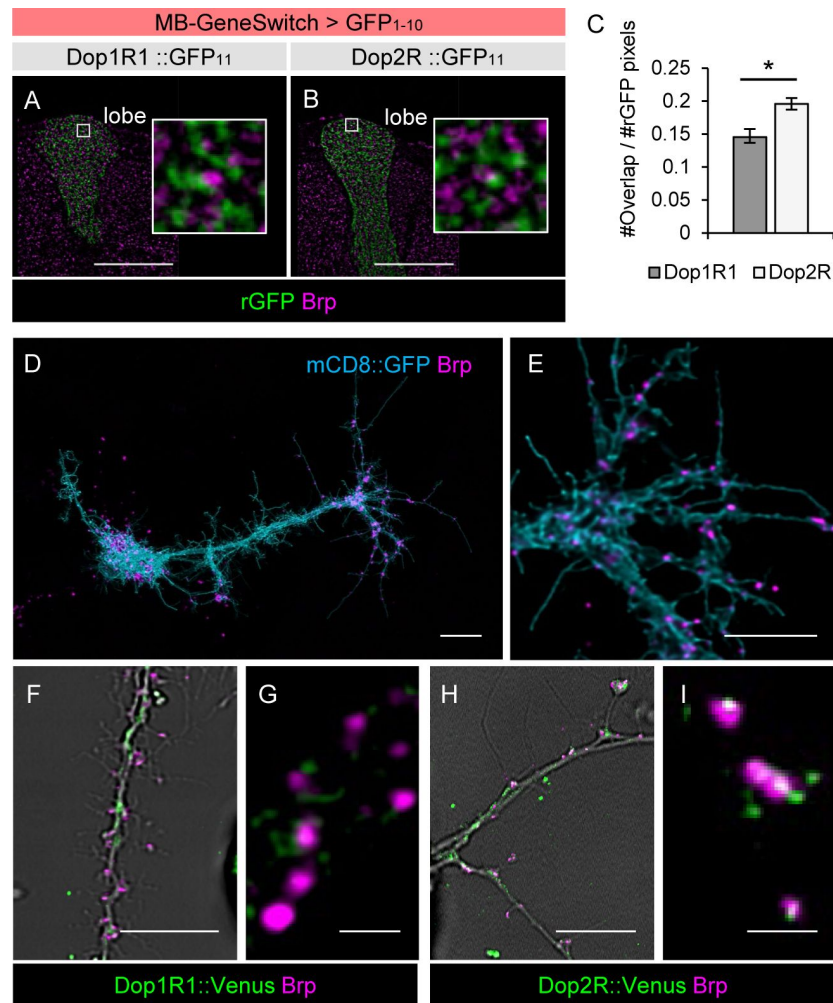


Figure 4.

Presynaptic localization of Dop1R1 and Dop2R in Kenyon Cells and giant neurons.

(A and B) Double labelling of dopamine receptors (green) and Brp (magenta). Single focal slice at the tip of the vertical lobe. Insertions are the magnified images of white squares in the main panel.

(C) Colocalization analysis showed that Dop2R had significantly higher colocalization with Brp than Dop1R1 ($p = 0.0259$, Mann-Whitney U test, $N = 6$). Error bars; SEM.

(D-E) Punctate Brp expression in a giant neuron culture differentiated from cytokinesis-arrested neuroblasts of *OK371-GAL4/UAS-mCD8::GFP* embryos. Aggregated Brp condensates (magenta) were observed in the neurite terminals of the cells marked with mCD8::GFP (cyan) in (E).

(F-I) Double labelling of dopamine receptors (green) and the AZs (magenta). *Dop1R1::Venus* (F and G) or *Dop2R::Venus* (H and I) was crossed with *Brp::SNAP*.

Scale bars, 20 μm (A and B), 10 μm (D, F and H), 5 μm (E), 1 μm (G and I).

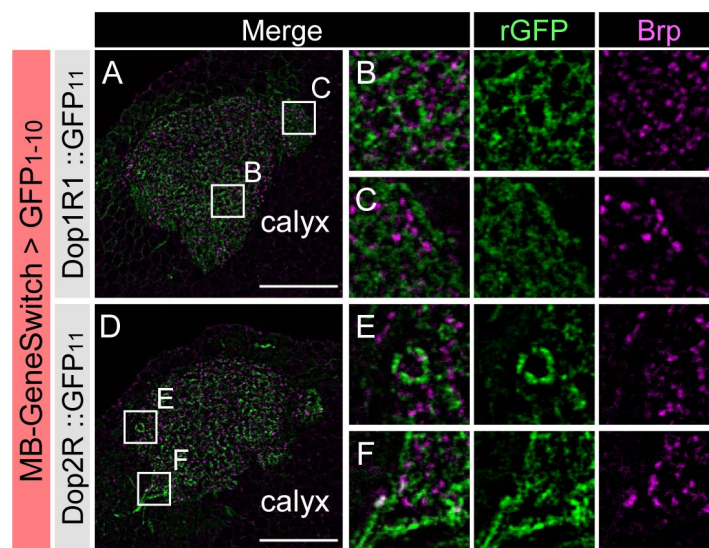


Figure 4– figure supplement 1.

Dop2R and Dop1R1 at the presynaptic sites of Kenyon Cells in the calyx.

(A-F) Double labelling of dopamine receptors (green) and Brp (magenta). From the single focal slice of the calyx (A and D), white squares including dendritic claws (B and E) and proximal dendrites (E and F) are magnified. Scalebars, 20 μ m (A and D).

5H-5J [5J](#)). Higher magnification revealed the accumulation of both Dop1R1 and Dop2R in the boutons (**Figure 5K** [5K](#) and **5L** [5L](#)). The presynaptic localization of Dop1R1 and Dop2R in DANs suggests that both receptors mediate feedback regulation.

State-dependent and bidirectional modulation of dopamine receptor expression in the PAM and PPL1 DANs

The activity of MB-projecting DANs is reported to be dynamic and sensitive to feeding states (Ichinose et al., 2017 [17](#); Liu et al., 2012 [12](#); Plaçais & Preat, 2013 [13](#); Senapati et al., 2019 [19](#); Siju et al., 2020 [20](#); Tsao et al., 2018 [18](#); Yamagata et al., 2016 [16](#)). We therefore examined if starvation alters the protein expression of Dop1R1 and Dop2R in these MB-projecting DANs (**Figure 6** [6](#)). We found a significant elevation of Dop1R1 in the terminals of PAM-γ5 upon starvation for 10 hours or longer (**Figure 6E** [6E](#)). In contrast, starvation did not increase Dop2R in PAM-γ5 but rather tended to decrease, if at all (**Figure 6E** [6E](#)). We found similar starvation-dependent changes in Dop1R1 and Dop2R levels in other PAM neurons (**Figure 6 – figure supplement 1A-H** [1A-H](#)). These results together suggest that starvation enhances presynaptic dopamine signaling in the reward-related PAM neurons by shifting the balance of Dop1R1 and Dop2R.

DANs of the PAM and PPL1 clusters exert distinct, largely opposite behavioral functions (Claridge-Chang et al., 2009 [9](#); Liu et al., 2012 [12](#)). Therefore, plasticity in the PPL1 neurons may differ from that in the PAM neurons. To test this hypothesis, we examined the starvation-dependent change in Dop1R1 and Dop2R protein expression in the PPL1 neurons. To this end, we visualized GFP₁₁-tagged receptors by expressing GFP₁₋₁₀ using *TH-GAL4* (Friggi-Grelin et al., 2003 [3](#)). We detected Dop2R proteins in all MB projections of the PPL1 neurons, whereas Dop1R1 proteins were only detectable in the terminals of the PPL1-α3 neuron (**Figure 6F** [6F](#) and **6G** [6G](#)). Strikingly, the starvation-induced changes in the PPL1 neurons were opposite to those in the PAM: Dop2R, but not Dop1R1, significantly increased in the α3 compartment (**Figure 6J** [6J](#)), and the same tendency was observed in the other PPL1 neurons (**Figure 6 – figure supplement 1I-L** [1I-L](#)). Taken together, starvation induces opposite responses of dopamine receptor expression in the PPL1 and PAM DANs, and we propose that these changes shift the balance of dopamine output from these subsets (**Figure 7A** [7A](#)).

The starvation-induced increase of Dop2R in the PPL1 terminals suggests enhanced negative feedback (Krashes et al., 2009 [9](#)). If so, loss of Dop2R would impair appetitive memory in starved flies, since inhibition of PPL1 output is necessary for appetitive memory expression (Krashes et al., 2009 [9](#); Pavlovsky et al., 2018 [18](#)) (**Figure 7B** [7B](#)). To test this hypothesis, we examined appetitive olfactory associative memory by transgenic knockdown of Dop2R specifically in the PPL1 cluster neurons using *MB504B-GAL4* (Vogt et al., 2014 [14](#)). Indeed, appetitive memory was impaired (**Figure 7C** [7C](#)), demonstrating the significance of the negative feedback regulation of the PPL1 in appetitive behavior expression. These results are in line with the state-dependent changes in the physiology of these DANs (Siju et al., 2020 [20](#); Tsao et al., 2018 [18](#)). Taken together, the opposite presynaptic modulation of the PPL1 and PAM neurons through the antagonistic dopamine autoreceptors determines the balance of the MB output to control appetitive behaviors.

Discussion

The present study demonstrated the expression, subcellular localization, and dynamics of endogenous dopamine receptors, specifically Dop1R1 and Dop2R, in the MB circuit of the fly brain. Previously, intronic insertions of fluorescent tags have been used to label endogenous proteins in a cell-type specific manner (Fendl et al., 2020 [20](#); R. Kamiyama et al., 2021 [21](#); Williams et al., 2019 [19](#)). CRISPR/Cas9-based insertion is advantageous in terms of flexibility in fusion sites compared to

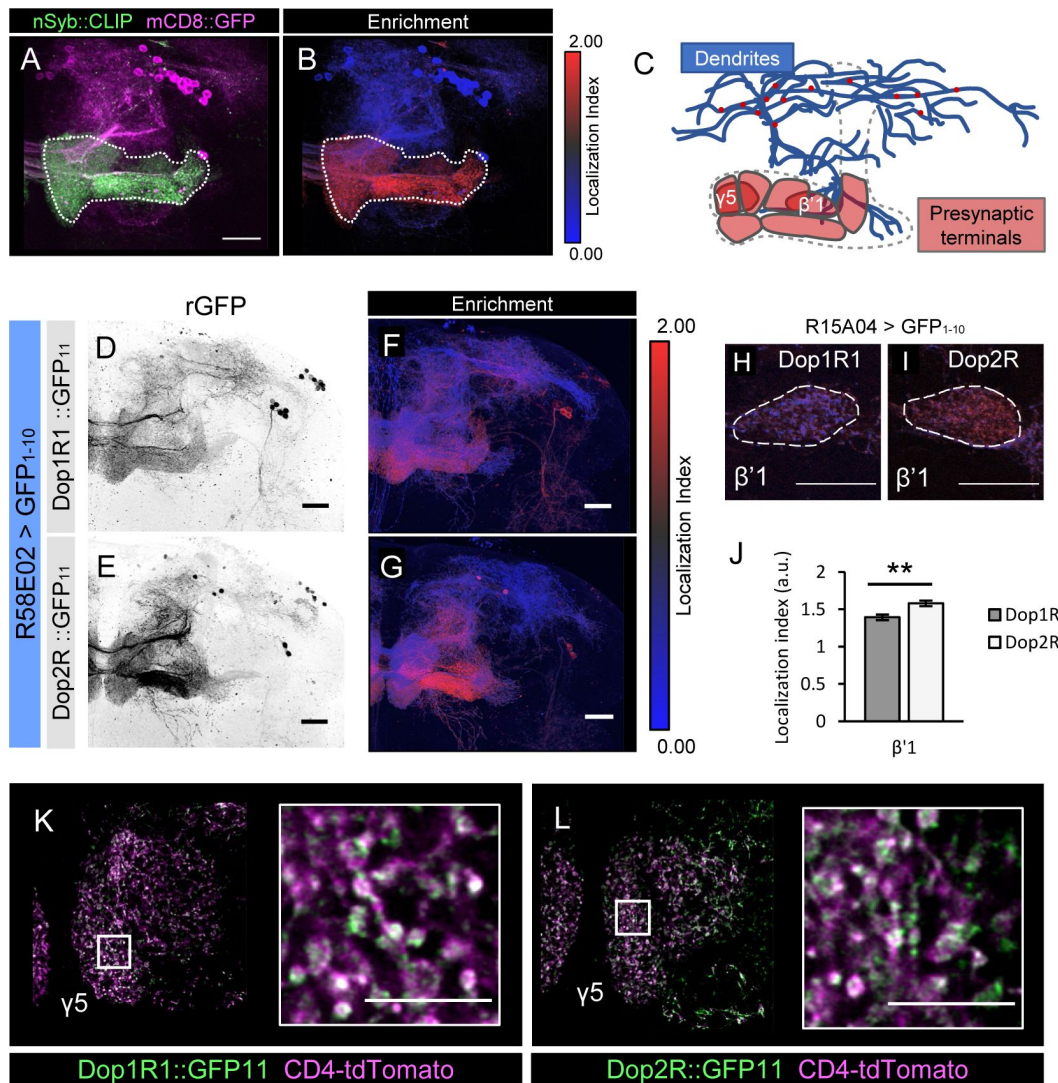


Figure 5.

Subcellular localization of Dop1R1 and Dop2R in dopamine neurons.

(A-B) Maximum-intensity projection image showing the distribution of presynaptic sites in the PAM neurons. (A) *R58E02-GAL4* was used to express *UAS-mCD8::GFP* (magenta) and *UAS-nSyb-CLIP* (magenta). (B) Visualization by LI showing enrichment of *nSyb* signals in the lobe projection of the PAM neurons.

(C) Illustrated projection pattern of the PAM neurons. Red puncta indicate the sparse distribution of presynaptic sites in dendrites.

(D-L) Subcellular localization of GFP₁₁-tagged Dop1R1 and Dop2R in the PAM neurons. *R58E02-GAL4* was used to express *UAS-GFP₁₋₁₀* and *UAS-CD4::tdTomato* in the PAM neurons.

(D and E) Reconstituted GFP signals of Dop1R1::GFP₁₁ (D) and Dop2R::GFP₁₁ (E) in the PAM neurons. Maximum-intensity projections of the left hemisphere including the MB lobe, SMP, SIP and SLP (superior medial, intermediate, and lateral protocerebrum, respectively). (F and G) Stronger presynaptic enrichment of Dop2R (G) than that of Dop1R1 (F) in the PAM dopamine neurons visualized by LI. β'1 compartment is magnified in the insertion image.

(H-J) *R15A04-GAL4* was used to measure LI in the PAM-β'1 neuron. (H and I) Presynaptic terminals of the PAM-β'1 neurons are shown (dashed line). (H) Mean LI for Dop1R1 and Dop2R in β'1 ($p = 0.0056$, Mann-Whitney U test, $N = 9$). Error bars; SEM.

(K and L) A single optical slice of γ5 compartment in the MB lobe obtained using Airyscan. Merged image of reconstituted GFP (green) and CD4::tdTomato (magenta). Insertions are the magnified images of the presynaptic boutons of PAM-γ5 (white squares).

Scale bars, 20 μm (A-B, D-I), 5 μm (K and L).

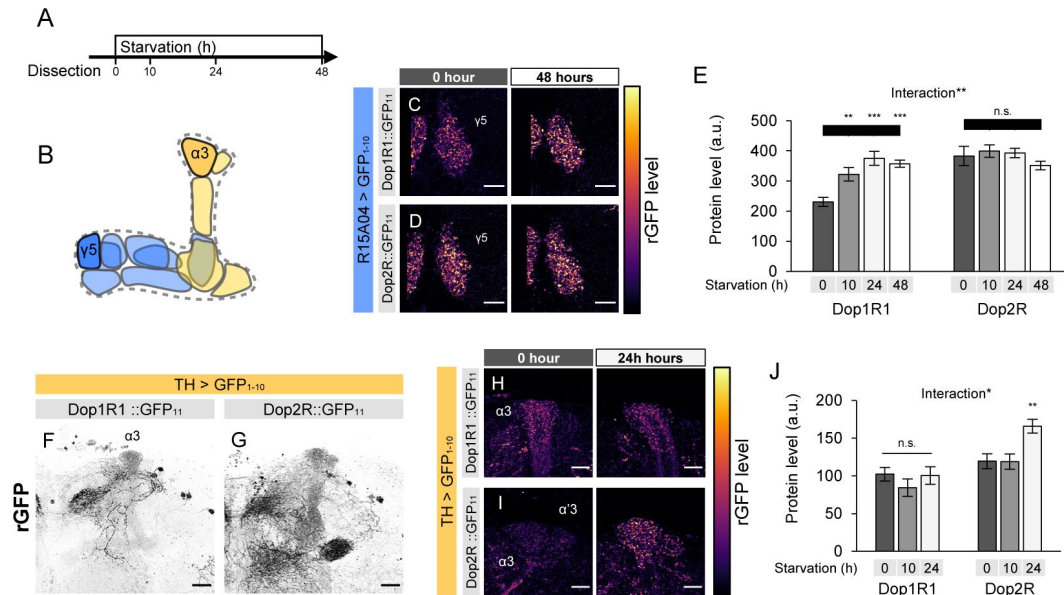


Figure 6.

Bidirectional modification of dopamine receptor expression in dopamine neurons.

(A) Schematic diagram of the starvation protocol. (B) Schematic illustration of the MB projection of the PAM and PPL1 dopamine neurons marked by *R15A04-GAL4* and *TH-GAL4*. (C and D) Dop1R (C) and Dop2R (D) in the presynaptic terminals of PAM-γ5 before and after 48 hours of starvation.

(E) Quantification of dopamine receptor levels in the presynaptic terminals of PAM-γ5 after 0, 10, 24 and 48 hours of starvation ($n = 6-13$).

(F and G) Reconstituted GFP signals of Dop1R1::GFP₁₁ (F) and Dop2R::GFP₁₁ (G) in the PPL1 neurons. In the MB projections of the PPL1 neurons, Dop1R1 is expressed in only the α3 compartment (F). Dop2R is expressed in all MB projections (G). Maximum-intensity projections of the MB lobe.

(H and I) Dop1R (H) and Dop2R (I) in the presynaptic terminals of PPL1-α3 before and after 24 hours of starvation.

(J) Quantification of the dopamine receptor levels in the presynaptic terminals of PPL1-α3 after 0, 10 and 24 hours of starvation ($n = 7-10$).

Interaction effects between genotypes and starvation time on protein levels were tested by Two-way ANOVA (E and J).

Scale bar, 10 μm (C, D, H and I), 20 μm (F and G).

Error bars; SEM (E and J). * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, n.s. = not significant.

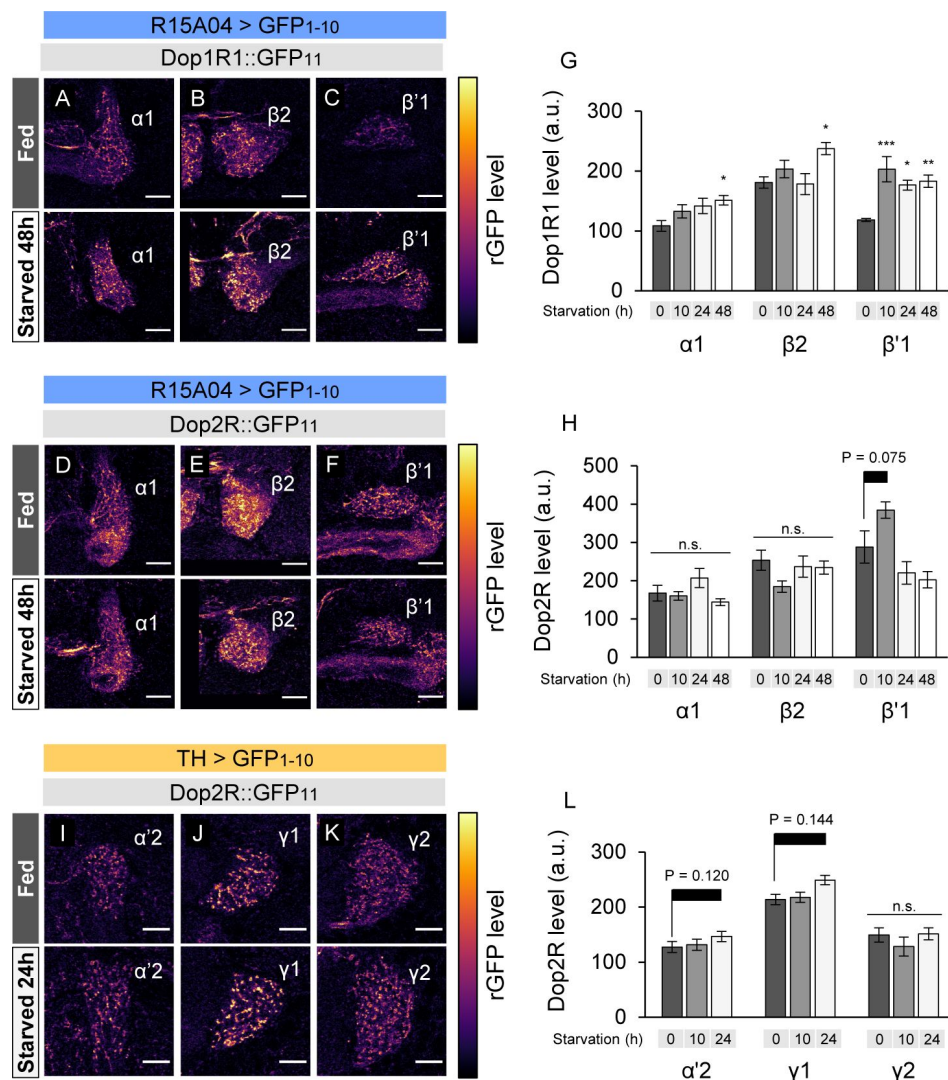


Figure 6– figure supplement 1.

Starvation-dependent change of dopamine receptors in PAM and PPL1.

(A-F) Dop1R (A-C) and Dop2R (D-F) in the presynaptic terminals of PAM- $\alpha 1$ (A and D), PAM- $\beta 2$ (B and E) and PAM- $\beta'1$ (C and F) before and after 48 hours of starvation.

(G and H) Quantification of dopamine receptor levels in the presynaptic terminals of the PAM neurons after 0, 10, 24 and 48 hours of starvation ($n = 6-13$).

(I-K) Dop2R in the presynaptic terminals of PPL1- $\alpha'2$ (I), PPL1- $\gamma 1$ pedc (J) and PPL1- $\gamma 2$ (K) before and after 24 hours of starvation.

(L) Quantification of Dop2R levels in the presynaptic terminals of the PPL1 neurons after 0, 10 and 24 hours of starvation ($n = 7-10$).

Scale bar, 10 μm (A-F, I-K). Error bars; SEM (G, H and L). * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, n.s. not significant.

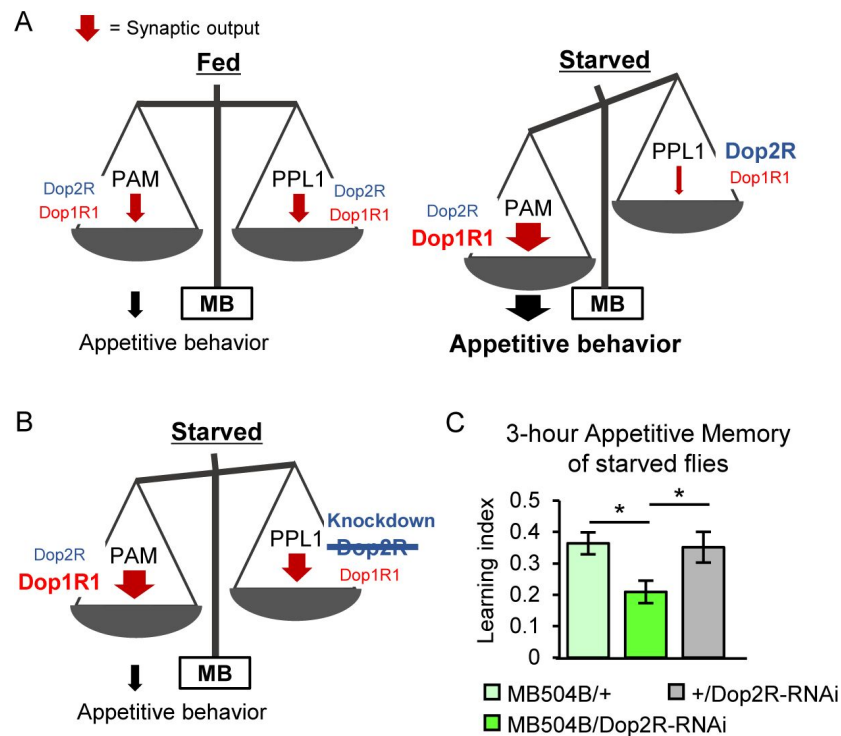


Figure 7.

The dual dopaminergic feedback regulating starved-state dependent expression of appetitive behavior.

(A) A working model showing the role of the dual dopaminergic feedback regulation. In starved state, increased Dop1R1 in PAM neurons and increased Dop2R in PPL1 neurons changes the balance between the synaptic outputs from these DANs to favor appetitive behavior.

(B) According to the model, loss of Dop2R in PPL1 upregulates output from PPL1 to attenuate appetitive behavior in starved flies.

(C) Knockdown of Dop2R in the PPL1 neurons by *MB504B-GAL4* reduced 3-hour appetitive memory performance (n=14-15). Error bars; SEM. * $p < 0.05$.

intronic insertions (Fendl et al., 2020 [DOI](#); R. Kamiyama et al., 2021 [DOI](#); Williams et al., 2019 [DOI](#)). This enables the C-terminal tagging of dopamine receptors that do not perturb protein localization and function (Kondo et al., 2020 [DOI](#)).

The split-GFP reconstitution strategy enables sensitive detection of low-abundant endogenous proteins by the tandem multimerization of the GFP₁₁ tags, as shown in other epitope tags such as SunTag and smFPs (Tanenbaum et al., 2014 [DOI](#); Viswanathan et al., 2015 [DOI](#)). The background fluorescence of the split-GFP fragments was practically negligible. For example, Dop1R1-rGFP signal was observed in fewer cells than GFP₁₋₁₀-expressing PPL1 neurons as expected from the gene expression pattern (D. Kamiyama et al., 2016 [DOI](#); Kondo et al., 2020 [DOI](#)) (**Figure 6** [DOI](#)). This approach can thus be applicable to monitor localization and dynamics of endogenous proteins of low abundance.

To quantify subcellular receptor localization, we here devised LI, which measures the relative enrichment of tagged proteins over the membrane markers. Quantitative localization analysis with LI revealed fine distinctions in presynaptic enrichment between Dop1R1 and Dop2R (**figure 3** [DOI](#) and **5** [DOI](#)). Dop2R showed a stronger association with AZ, while Dop1R1 was distributed more broadly along axons and terminals (**Figure 4** [DOI](#)). Similar but distinct subcellular enrichment of Dop1R1 and Dop2R therefore suggests joint action in intricate regulation of presynaptic dopamine signaling.

Spatial regulation of dopamine signaling through distinct receptor subtypes

How do Dop1R1 and Dop2R function in presynaptic terminals? Differential AZ enrichment suggests differential contributions to the spatial regulation of dopamine response. Dopamine release is not confined to the synaptic cleft, but involves extracellular diffusion (Rice & Cragg, 2008 [DOI](#)) and extrasynaptic release (Liu & Kaeser, 2019 [DOI](#)). Such volume transmission can increase the number of target synapses by 10 times (Li et al., 2020 [DOI](#); Takemura et al., 2017 [DOI](#)). The broader distribution of Dop1R1 thus enables to respond to the trace amount of extrasynaptic dopamine. On the other hand, the tight association of Dop2R with the AZ underscores the importance of reducing the noise of second messenger signaling likely through inhibiting activities of adenylyl cyclase and voltage-gated calcium channels (Beaulieu & Gainetdinov, 2011 [DOI](#); Hearn et al., 2002 [DOI](#)). These second messengers, cAMP and Ca²⁺, were reported to regulate distinct aspects of presynaptic functions including release probability and vesicle recycling (Ehmann et al., 2018 [DOI](#); Kuromi & Kidokoro, 2000 [DOI](#), 2005 [DOI](#)). The spatial configuration of Dop1R1 and Dop2R may thus contribute to the sensitivity and precision in dopaminergic modulation of the synaptic vesicle cycle.

The dual autoreceptor system may shape dopamine release

Presynaptic localization of Dop1R1 and Dop2R in DANs strongly suggests their functions as autoreceptors. Reported functions of inhibitory dopamine autoreceptors range from negative regulations of the amplitude and kinetics of dopamine transmission, decreasing dopamine synthesis, to altering the uptake of released dopamine (Ford, 2014 [DOI](#); Shin & Venton, 2022 [DOI](#); Vickrey & Venton, 2011 [DOI](#)). Considering tight presynaptic Dop2R enrichment in the PAM and PPL1 DANs, such negative feedback should prevent overactivation of second messenger signaling at the AZ, where the ligand concentrations should be highest. In contrast, the presence of Dop1R1 in DAN terminals was unexpected, adding another layer of presynaptic regulations in dopamine release. Although it must be functionally verified if presynaptic Dop1R1 acts as autoreceptors in the PAM and PPL1 DANs, it should provide feedback for the dopamine release. This positive feedback would be particularly necessary in the regions where extracellular dopamine concentrations are low. Indeed, presynaptic Dop1R1 was undetectable in the PPL1 DNAs, which were reported to have high spontaneous activities (Feng et al., 2021; Plačaiš & Preat, 2013 [DOI](#)). Furthermore, we noticed

that compositions of presynaptic Dop1R1 and Dop2R vary among individual DANs (**Figure 5** [↗](#) and **6** [↗](#)). This dual autoreceptor system may thus fine-tune the amplitude and kinetics of dopamine release depending on the cellular function.

Dopamine receptor expression is reported to be associated with prolonged exposure to psychoactive substances, such as caffeine and ethanol (Andretic et al., 2008 [↗](#); Kanno et al., 2021 [↗](#); Kondo et al., 2020 [↗](#); Petrucci et al., 2018 [↗](#)). This study further showed starvation-dependent changes of Dop1R1 and Dop2R in DAN terminals (**Figure 7** [↗](#)). Strikingly, starvation responses of presynaptic Dop1R1 and Dop2R were differential depending on the DAN cell types (**Figure 7** [↗](#)). We therefore propose the state-dependent bidirectional modulation of the MB output through the dual dopamine autoreceptor system in the PAM and PPL1 neurons (**Figure 7A** [↗](#)). Starvation changes the dual autoreceptor system to put more weight on the PAM output over PPL1 to control the expression of appetitive behavior (**Figure 7A** [↗](#)). This shifted balance explains the state-dependent changes in the presynaptic activity of these two clusters of DANs and recruited MBONs for appetitive behavior (Aso, Sitaraman, et al., 2014; Ichinose et al., 2021 [↗](#); Landayan et al., 2018 [↗](#); Oswald et al., 2015 [↗](#); Siju et al., 2020 [↗](#); Tsao et al., 2018 [↗](#)).

Materials and Methods

Resource tables

Strain	Source	Identifier
Antibodies		
Mouse anti-TH	ImmunoStar Inc., Hudson, WI, USA	Cat# 22941
Mouse anti-Brp	Developmental Studies Hybridoma Bank, Univ. Iowa	nc82
Goat anti-mouse AlexaFluor 633	Invitrogen, Waltham, MA, USA	Cat# A-21052
Drosophila strain		
<i>Dop1R1-T2A-GAL4</i>	Kondo et al., 2020	N/A
<i>Dop2R-T2A-GAL4</i>	Kondo et al., 2020	N/A
<i>Dop1R1-T2A-LexA</i>	Kondo et al., 2020	N/A
<i>Dop2R-T2A-LexA</i>	Kondo et al., 2020	N/A
<i>Dop1R1::7xGFP11</i>	Kondo et al., 2020	N/A
<i>Dop2R::7xGFP11</i>	Kondo et al., 2020	N/A
<i>DopEcR::7xGFP11</i>	Kondo et al., 2020	N/A
<i>Dop1R1::Venus</i>	Kondo et al., 2020	N/A
<i>Dop2R::Venus</i>	Kondo et al., 2020	N/A
<i>UAS-GFP1-10</i>	Kondo et al., 2020	N/A
<i>UAS-mCD8::GFP</i>	Pfeiffer et al., 2012	BDSC #32194
<i>lexAop-rCD2::GFP</i>	Miyamoto et al., 2012	N/A
<i>UAS-CD4::tdTomato</i>	C. Han et al., 2011	BDSC #35841
<i>UAS-nSyb-CLIP</i>	Kohl et al., 2014	BDSC #58398
<i>Brp::SNAP</i>	Kohl et al., 2014	BDSC #58397
<i>MB-GeneSwitch</i>	Mao et al., 2004	BDSC #81013
<i>R15A04-GAL4</i>	Pfeiffer et al., 2008	BDSC #48671
<i>R58E02-GAL4</i>	Pfeiffer et al., 2008	BDSC #41347
<i>MB504B</i>	Pfeiffer et al., 2010	BDSC #68329
<i>OK371-GAL4</i>	Mahr & Aberle, 2006	BDSC #26160
<i>UAS-Dop2R-RNAi</i>	Perkins et al., 2015	BDSC #50621
Software		
Fiji	Schindelin et al., 2012, http://fiji.sc	RRID: SCR_002285
GraphPad Prism 5	GraphPad Software	RRID: SCR_002798

Flies

Flies were raised on standard cornmeal food at 25°C under a 12:12 h light-dark cycle (ZT0 at 8 AM) for all experiments. The GAL4-UAS system was used to express the transgenes of interest in specific neuron subtypes. Flies carrying GAL4 were crossed to another strain carrying UAS reporters, and F1 progenies were used in the experiments. To visualize GFP₁₁-tagged dopamine receptors in the specific cell-types, female fly strains carrying *UAS-CD4::tdTomato*, *UAS-GFP₁₋₁₀* and GAL4 driver was crossed to male fly strains carrying *Dop1R1::GFP₁₁* or *Dop2R::GFP₁₁* and F1

progenies were used. To make the giant neuron culture, embryos from *Brp::SNAP* or the F1 progeny of *OK371-GAL4* (BDSC #26160) crossed with *Brp::SNAP, UAS-mCD8::GFP, UAS-nSyb-CLIP* flies were used. For the fly lines used in our manuscript, see the Resource Tables.

Brain dissection and Immunohistochemistry

Male flies with the specific genotype were sorted under CO₂ anesthesia and kept in a food vial for recovery at least 24 h prior to the dissection. Fly brains were dissected 3-7 days after eclosion in ice-cold Phosphate-buffered saline (PBS). After dissection, brains were kept in ice-cold PBS with 2% paraformaldehyde (PFA) for up to 30 min. For fixation, brains were incubated in 2 % PFA in PBS for 1 hour at room temperature. Fixed brains were washed in PBS with 0.1% Triton X-100 (PBT) for 10 min 3 times. For rGFP imaging, fixed brains were mounted in SeeDB2 (Ke et al., 2016), and native fluorescence was imaged without signal amplification.

For chemical tagging reaction (Figure 5A and 5B), brains were incubated in PBT (0.3 % Triton X-100 in PBS) containing CLIP-Surface 547 substrate (1μM, S9233S; New England Biolabs Inc.) for 15 min at room temperature. Brains were washed in PBT for 10 min 3 times.

For immunohistochemistry, fixed brains were blocked in 3 % normal goat serum (NGS) for 30 min at room temperature. Brains were incubated in primary and secondary antibodies diluted in PBT with 1 % NGS over two nights at 4 °C, respectively. After each step, brains were washed three times in PBT for longer than 20 min at room temperature and mounted in 86% glycerol.

Following primary antibodies were used: mouse anti-TH (1:100; 22941; ImmunoStar Inc., Hudson, WI, USA), mouse anti-Brp (1:40; nc82; Developmental Studies Hybridoma Bank, Univ. Iowa). Secondary antibodies: Alexa Fluor 633 goat anti-mouse (1:400 for anti-TH, 1:200 for nc82; A-21052; Invitrogen, Waltham, MA, USA).

Fluorescent imaging

For image acquisition, whole-mount brains were scanned with the Olympus FV1200 confocal microscope with following objective lens; 20x oil (NA = 0.85, UPLSAPO20XO, Olympus), 30x silicone (NA = 1.05, UPLSAPO30XS, Olympus), 40x oil (NA = 1.3, UPLFLN40XO, Olympus), or 60x oil (NA = 1.42, PLAPON60XO, Olympus). Z-stack images were acquired with following voxel sizes; 0.468 x 0.468 x 0.84 μm³ for 20x oil (Figure 1A and 1B), 0.099 x 0.099 x 0.44 μm³ for 30x silicone (Figure 1D-1H), 0.194 x 0.194 x 0.52 μm³ for 40x oil (Figure 3B-3G, 6C-6D, 6F-6I and Figure 6 – figure supplement 1 A-F, I-K), and 0.101 x 0.101 x 0.45 μm³ for 60x oil (Figure 4A, 4B and Figure 4 – figure supplement 1 A-F).

For high resolution imaging in Figure 5I-J, Airyscan imaging was performed on Zeiss LSM800 with 63x oil objective lens (NA = 1.40, Plan-Apochromat). A single focal slice was scanned with a pixel size of 0.0425 x 0.0425 μm².

Image analysis

All image analysis were conducted on Fiji (Schindelin et al., 2012).

To visualize the subcellular localization of dopamine receptors in defined neurons, we devised the localization index (LI). Relative receptor density was calculated in each voxel by dividing the receptor (reconstituted GFP) signals by the corresponding membrane signals (CD4::tdTomato). To set ROI, any voxels devoid of membrane signal were censored. The local density was normalized by the mean values in the ROI. For the visual representation, the reference membrane signals were colored according to the normalized LI (Figure 2C, middle). To quantify mean LI in each subcellular region (Figure 3H), ROI was manually set at single focal slice that include the largest area of interested region.

Three-dimensional iterative deconvolution with the Richardson-Lucy algorithm was applied to some stack images (**Figure 4A-B**, **Figure 4 – figure supplement 1**). To generate the theoretical PSF for deconvolution, the ImageJ plug-in diffraction PSF 3D was used. Deconvolved images were used for following colocalization analysis.

To analyze the colocalization of dopamine receptors and Brp (**Figure 4C**), both channels were binarized with the auto-thresholding on Fiji. The ratio of overlap between rGFP and Brp signals over rGFP signals was calculated. Square ROI (10.1 x 10.1 μm , 100 x 100 pixel) at single focal slice was sampled within the tip of the vertical lobes for colocalization analysis.

Embryonic giant neuron culture

Multinucleated giant neurons from neuroblasts were generated as described previously (Wu et al., 1990). Briefly, the interior of a gastrula from stage 6/7 embryo was extracted with a glass micropipette and dispersed into a drop of culture medium ($\sim 40 \mu\text{m}$) on an uncoated coverslip. The culture medium contained 80% *Drosophila* Schneider's insect medium (S0146-100ML, Merck KGaA, Darmstadt, Germany) and 20% fetal bovine serum (F2442-100ML, Merck KGaA, Darmstadt, Germany), with the addition of 50 $\mu\text{g/ml}$ streptomycin, 50 U/ml penicillin (all from Sigma, St. Louis, MO, USA) and cytochalasin B (CCB; 2 $\mu\text{g/ml}$; Sigma, St. Louis, MO, USA). CCB was removed by washing with CCB-free medium 1 d after plating. All cultures were grown in a humidified chamber.

To label Brp::SNAP, cells were incubated in fluorescent SNAP substrate diluted in the culture medium (1:5000, SNAP-Cell 647-SiR, S9102S, Ipswich, MA, USA) for half an hour at RT. For double labeling with nSyb-CLIP, cells were first incubated in fluorescent CLIP substrate diluted in the culture medium (1:5000, CLIP-Cell™ TMR-Star, S9219S, Ipswich, MA, USA) for half an hour at RT. Subsequently, the same cells were labeled by the SNAP substrate to minimize the crosstalk. After the incubation, cells were washed with the culture medium and subjected to confocal scanning with Leica TCS SP8 microscopy equipped with a 40x oil immersion objective (HC PL APO 40x/1.30 Oil PH3 CS2, Leica). Acquired images were then processed with the LIGHTNING software.

Behavioral assays

The experimental protocols were as described previously (Ichinose and Tanimoto, 2016). For appetitive conditioning, a group of approximately 50 flies in a training tube alternately received 3-octanol (3OCT; Merck) and 4-methylcyclohexanol (4MCH; Sigma-Aldrich), for 1 minute in a constant air flow with or without reward with an interval of 1 minute between the two odor presentations. These odorants were diluted to 1.2% and 2% with paraffin wax (Sigma-Aldrich), respectively. Dried 2 M sucrose (Sigma-Aldrich) on a piece of filter paper was used as the reward. Flies were starved in the presence of wet tissue paper for 24 hours before appetitive conditioning. For testing, flies were given a choice between CS+ and CS-, which distribution in the plexiglass T-maze (Ichinose and Tanimoto, 2016) was video-recorded one frame per second for 2 minutes with the CMOS cameras (GS3-U3-51S5M, FLIR) with infrared LED illumination (Ichinose & Tanimoto, 2016). Flies were forced to stay on the floor by applying Fluon (Insect-a-Slip PTFE30, BioQuip Products) on the side and top of T-maze arms.

Learning index was then calculated by counting the number of flies in each arm using ImageJ macro as described previously (Ichinose and Tanimoto, 2016) with the following formula (Tempel et al., 1983):

$$\text{Learning index} = \frac{\#_{\text{CS-}} - \#_{\text{CS+}}}{\#_{\text{CS+}} + \#_{\text{CS-}}}$$

where $\#_{\text{CS+}}$ and $\#_{\text{CS-}}$ imply the number of flies in the CS+ and the CS-arms, respectively. Learning index was calculated for every second and was averaged for the last 60 s of the 120 s test session. An average of a pair of reciprocally trained groups was used as a single data point.

Statistics

For multiple comparison (**Figure 6** [↗](#) and **Figure 6 – figure supplement 1** [↗](#)), statistics were performed by Prism5 (GraphPad). Data were always first tested for normality (Shapiro-Wilk test) and for homoscedasticity (Spearman's test or Bartlett's test). If these assumptions were not violated, parametric tests (one-way ANOVA, followed by Dunnett's post hoc pairwise test) were applied. If data did not suffice the assumptions for parametric tests, nonparametric tests (Kruskal-Wallis, followed by Dunn's post hoc pairwise test) were performed. For comparison of the two groups, Student's t test (**Figure 3H** [↗](#)) or Mann-Whitney U test (**Figure 4C** [↗](#), **5H** [↗](#)) was performed on R Statistical Software.

Bars and error bars represent means and SEM, respectively, in all figures. For all figures, significance corresponds to the following symbols: * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$; n.s. not significant.

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Author contributions

H.T. and S.H. conceived the project.

H.T. S.H. and N.Y. designed experiments and interpreted results.

S.K. and H.K. designed, generated, and provided transgenic flies.

H.T. and C-F. W. supervised experiments.

S.H., K.S. and N.Y. conducted experiments and analyzed data.

H.T., and S.H. drafted the manuscript.

All authors wrote and revised the manuscript.

All authors read and approved the final manuscript.

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National Centre for Biological Sciences, Tata Institute of Fundamental Research, Bangalore, India

Reviewer #1 (Public Review):

Summary:

This is an important and interesting study that uses the split-GFP approach. Localization of receptors and correlating them to function is important in understanding the circuit basis of behavior.

Strengths:

The split-GFP approach allows visualization of subcellular enrichment of dopamine receptors in the plasma membrane of GAL4-expressing neurons allowing for a high level of specificity.

The authors resolve the presynaptic localization of DopR1 and Dop2R, in "giant" *Drosophila* neurons differentiated from cytokinesis-arrested neuroblasts in culture as it is not clear in the lobes and calyx.

Starvation-induced opposite responses of dopamine receptor expression in the PPL1 and PAM DANs provide key insights into models of appetitive learning.

Starvation-induced increase in D2R allows for increased negative feedback that the authors test in D2R knockout flies where appetitive memory is diminished.

This dual autoreceptor system is an attractive model for how amplitude and kinetics of dopamine release can be fine-tuned and controlled depending on the cellular function and

this paper presents a good methodology to do it and a good system where the dynamics of dopamine release can be tested at the level of behavior.

Weaknesses:

LI measurements of Kenyon cells and lobes indicate that Dop2R was approximately twice as enriched in the lobe as the average density across the whole neuron, while the lobe enrichment of Dop1R1 was about 1.5 times the average, are these levels consistent during different times of the day and the state of the animal. How were these conditions controlled and how sensitive are receptor expression to the time of day of dissection, staining, etc.

The authors assume without discussion as to why and how presynaptic enrichment of these receptors is similar in giant neurons and MB.

Figures 1-3 show the expensive expression of receptors in alpha and beta lobes while Figure 5 focusses on PAM and localization in γ and β' projections of PAM leading to the conclusion that pre-synaptic dopamine neurons express these and have feedback regulation. Consistency between lobes or discussion of these differences is important to consider.

Receptor expression in any learning-related MBONs is not discussed, and it would be intriguing as how receptors are organized in those cells. Given that these PAMs input to both KCs and MBONs these will have to work in some coordination.

Although authors use the D2R enhancement post starvation to show that knocking down receptors eliminated appetitive memory, the knocking out is affecting multiple neurons within this circuit including PAMs and KCs. How does that account for the observed effect? Are those not important for appetitive learning?

The evidence for fine-tuning is completely based on receptor expression and one behavioral outcome which could result from many possibilities. It is not clear if this fine-tuning and presynaptic feedback regulation-based dopamine release is a clear possibility. Alternate hypotheses and outcomes could be considered in the model as it is not completely substantiated by data at least as presented.

<https://doi.org/10.7554/eLife.98358.1.sa1>

Reviewer #2 (Public Review):

Summary:

Hiramatsu et al. investigated how cognate neurotransmitter receptors with antagonizing downstream effects localize within neurons when co-expressed. They focus on mapping the localization of the dopaminergic Dop1R1 and Dop2R receptors, which correspond to the mammalian D1- and D2-like dopamine receptors, which have opposing effects on intracellular cAMP levels, in neurons of the *Drosophila* mushroom body (MB). To visualize specific receptors in single neuron types within the crowded MB neuropil, the authors use existing dopamine receptor alleles tagged with 7 copies of split GFP to target reconstitution of GFP tags only in the neurons of interest as a read-out of receptor localization. The authors show that both Dop1R1 and Dop2R, with differing degrees, are enriched in axonal compartments of both the Kenyon Cells cholinergic presynaptic inputs and in different dopamine neurons (DANs), which project axons to the MB. Co-localization studies of dopamine receptors with the presynaptic marker Brp suggest that Dop1R1 and, to a larger extent Dop2R, localize in the proximity of release sites. This localization pattern in DANs suggests that Dop1R1 and Dop2R work in dual-feedback regulation as autoreceptors. Finally, they provide evidence that the balance of Dop1R1 and Dop2R in the axons of two different

DAN populations is differentially modulated by starvation and that this regulation plays a role in regulating appetitive behaviors.

Strengths:

The authors use reconstitution of GFP fluorescence of split GFP tags knocked into the endogenous locus at the C-terminus of the dopamine receptors as a readout of dopamine receptor localization. This elegant approach preserves the endogenous transcriptional and post-transcriptional regulation of the receptor, which is essential for studies of protein localization.

The study focuses on mapping the localization of dopamine receptors in neurons of the mushroom body. This is an excellent choice of system to address the question posed in this study, as the neurons are well-studied, and their connections are carefully reconstructed in the mushroom body connectome. Furthermore, the role of this circuit in different behaviors and associative memory permits the linking of patterns of receptor localization to circuit function and resulting behavior. Because of these features, the authors can provide evidence that two antagonizing dopamine receptors can act as autoreceptors within the axonal compartment of MB innervating DANs. The differential regulation of the balance of the two receptors under starvation in two distinct DAN innervations provides evidence of the role that regulation of this balance can play in circuit function and behavioral output.

Weaknesses:

The approach of using endogenously tagged alleles to study localization is a strength of this study, but the authors do not provide sufficient evidence that the insertion of 7 copies of split GFP to the C terminus of the dopamine receptors does not interfere with the endogenous localization pattern or function. Both sets of tagged alleles (1X Venus and 7X split GFP tagged) were previously reported (Kondo et al., 2020), but only the 1X Venus tagged alleles were further functionally validated in assays of olfactory appetitive memory. Despite the smaller size of the 7X split-GFP array tag knocked into the same location as the 1X Venus tag, the reconstitution of 7 copies of GFP at the C terminus of the dopamine receptor, might substantially increase the molecular bulk at this site, potentially impeding the function of the receptor more significantly than the smaller, single Venus tag. The data presented by Kondo et al. 2020, is insufficient to conclude that the two alleles are equivalent.

The authors' conclusion that the receptors localize to presynaptic sites is weak. The analysis of the colocalization of the active zone marker Brp whole-brain staining with dopamine receptors labeled in specific neurons is insufficient to conclude that the receptors are localized at presynaptic sites. Given the highly crowded neuropil environment, the data cannot differentiate between the receptor localization postsynaptic to a dopamine release site or at a presynaptic site within the same neuron. The known distribution of presynaptic sites within the neurons analyzed in the study provides evidence that the receptors are enriched in axonal compartments, but co-labeling of presynaptic sites and receptors in the same neuron or super-resolution methods are needed to provide evidence of receptor localization at active zones. The data presented in Figures 5K-5L provides compelling evidence that the receptors localize to neuronal varicosities in DANs where the receptors could play a role as autoreceptors.

Given the highly crowded environment of the mushroom body neuropil, the analysis of dopamine receptor localization in Kenyon cells is not conclusive. The data is sufficient to conclude that the receptors are preferentially localizing to the axonal compartment of Kenyon cells, but co-localization with brain-wide Brp active zone immunostaining is not sufficient to determine if the receptor localizes juxtaposed to dopaminergic release sites, in proximity of release sites in Kenyon cells, or both.

